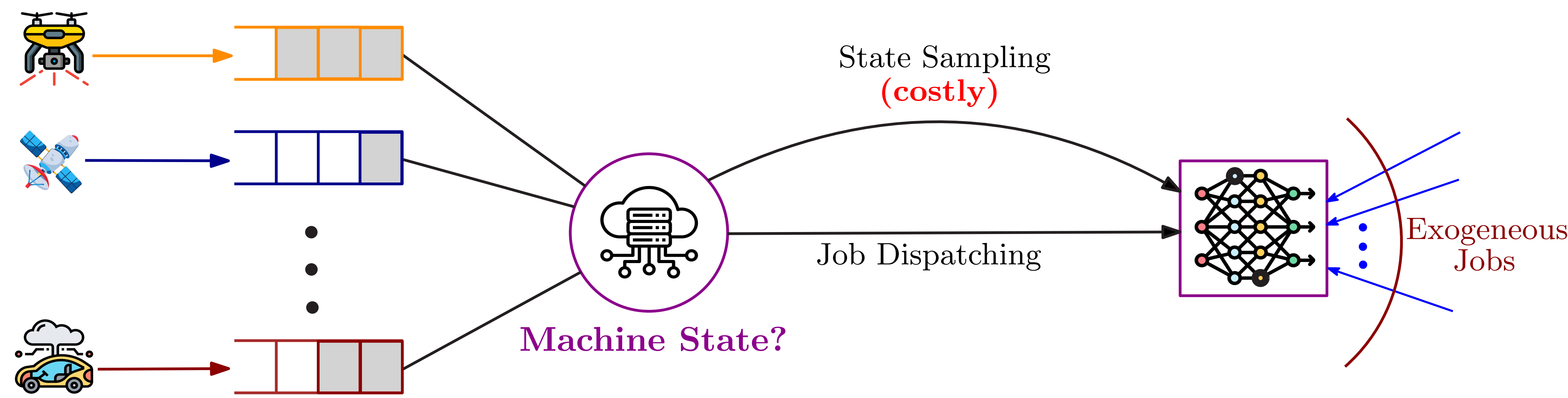


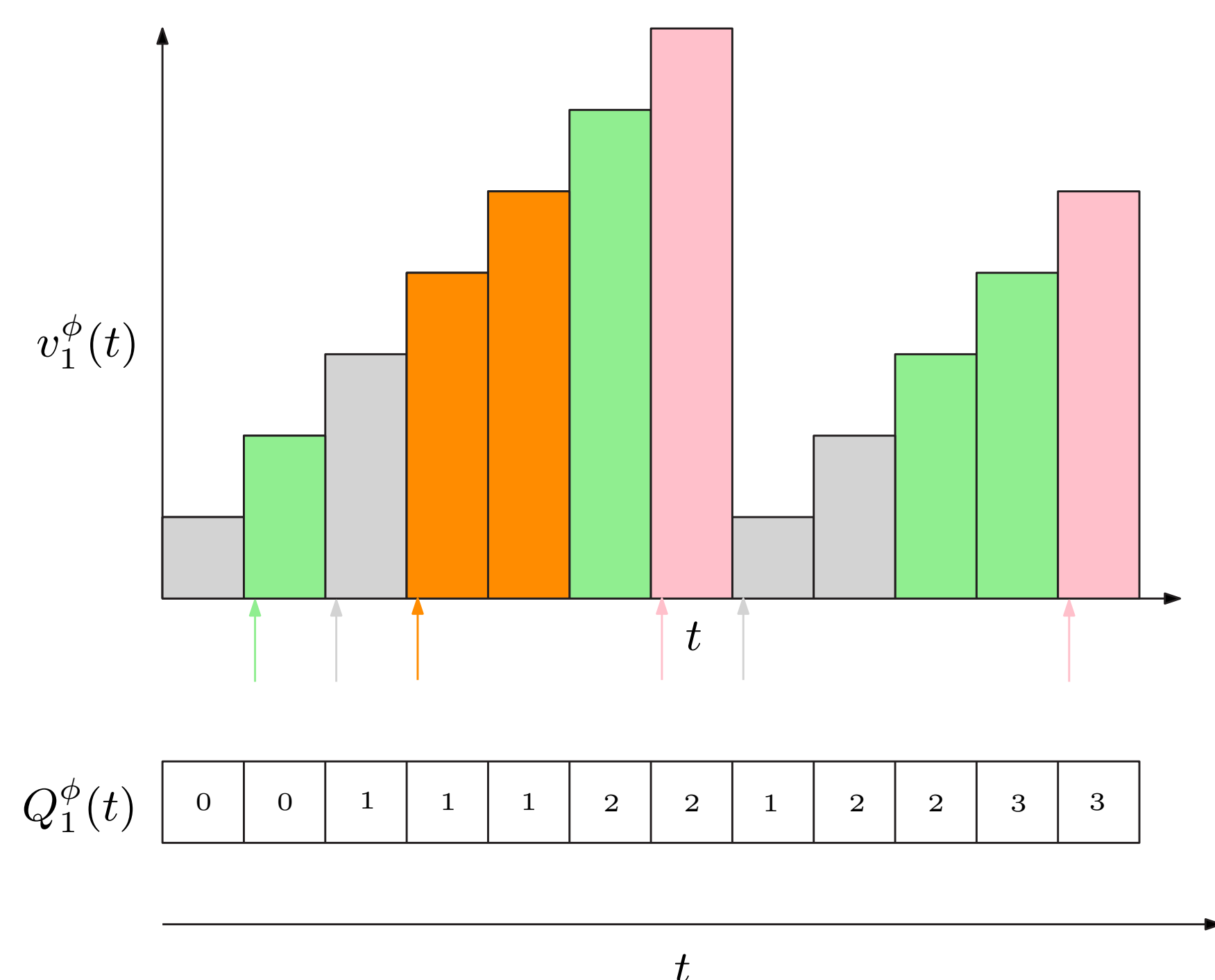
1. Core Problem

Dispatching jobs from a server to an edge machine whose immediate availability is **unknown** a priori. **Different** from conventional job offloading from users to a server.



2. Age of Job Completion (AoJC)

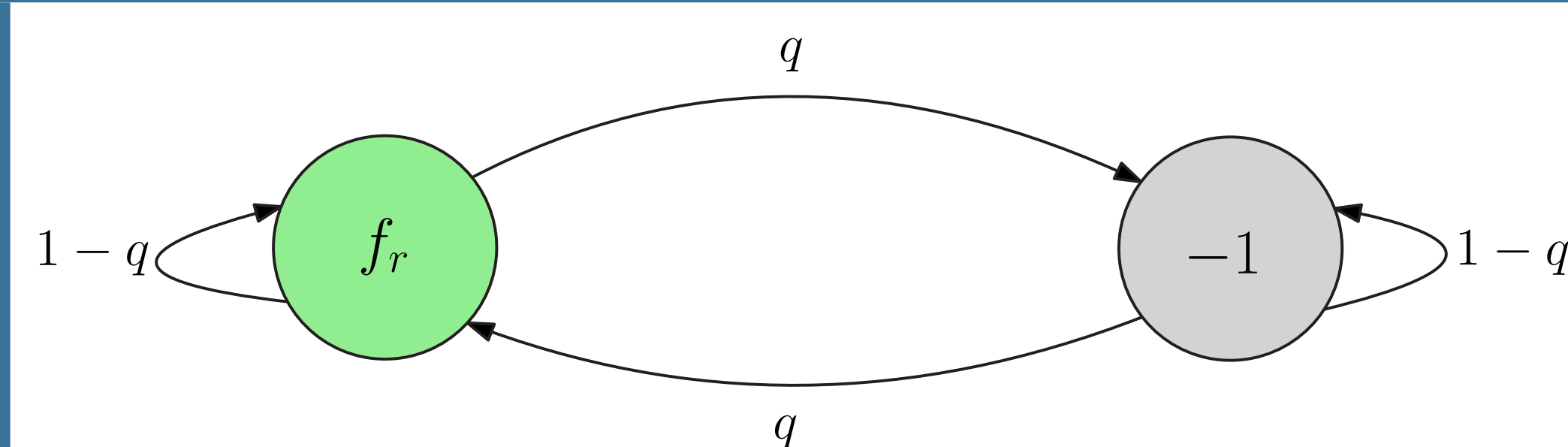
Definition: How long has it been since user i last got served?



Minimizing total average AoJC:

- prevents any single user from starving,
- keeps average inter-job completion time small,
- promotes frequent job completions,

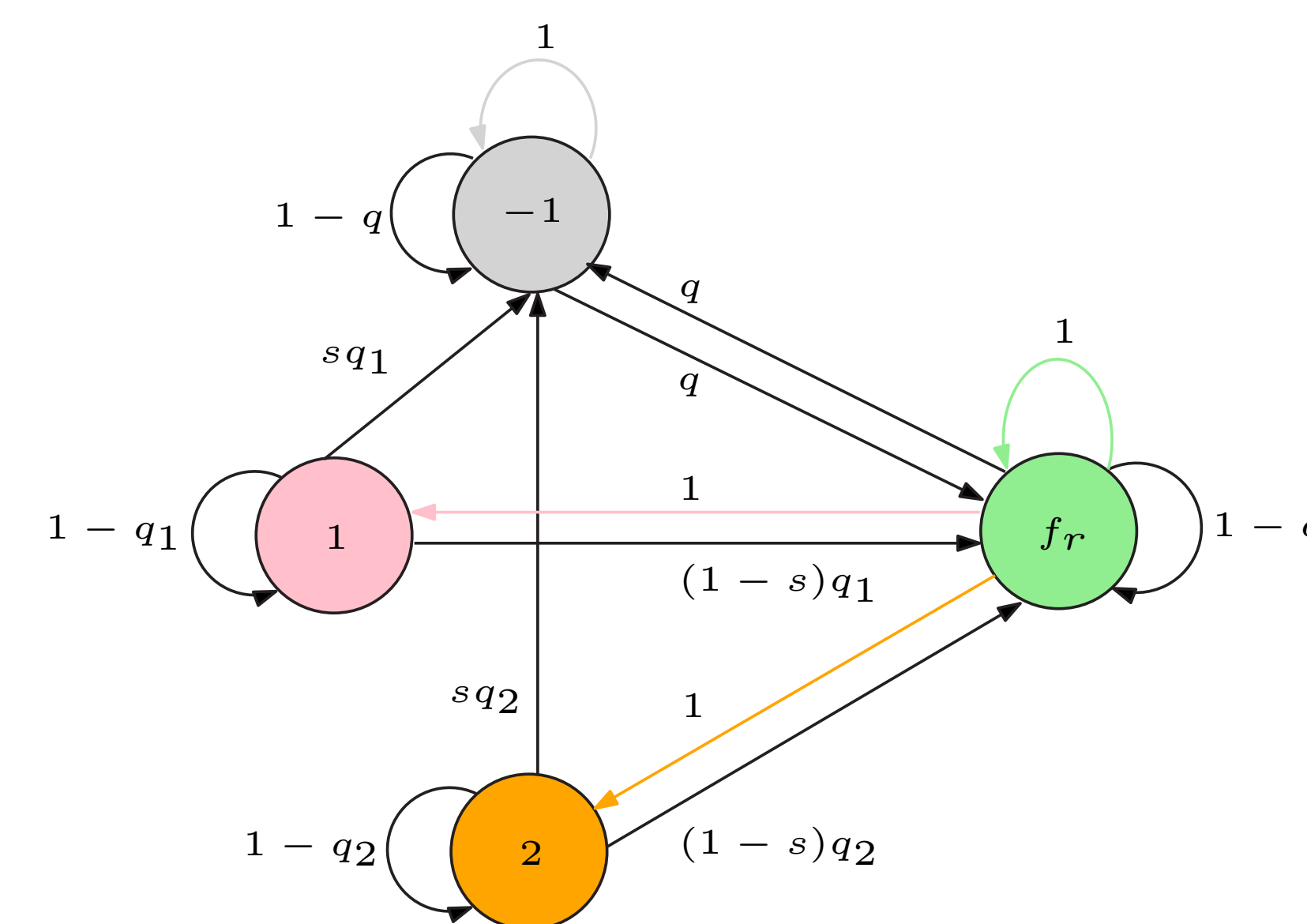
3. Tracking and Sampling



- Machine availability is unknown at the server.
- Sampling incurs a cost ($\mathbf{L}$), even if the machine is busy with an exogenous job.

4. System Model

- User i generates a new job in each slot with probability p_i .
- An example for $N = 2$ users:



- Sample: machine is free, but queues are empty.
- Sample: machine is free, schedule first user.
- Sample: machine is free, schedule second user.
- Sample: machine is internally busy.

- For a job assigned by the server:
 - completion information is received at the end of every slot,
 - service time follows a geometric distribution.
- The availability of the machine, disrupted by temporally correlated exogenous jobs (modeled as a Markov chain), is known to the server only through sampling.

5. Objective

Design joint sampling and scheduling policies that minimize total average AoJC and sampling cost while ensuring job-queue stability.

6. Proposed Policies and Queue Stability

Adaptive randomized policy:

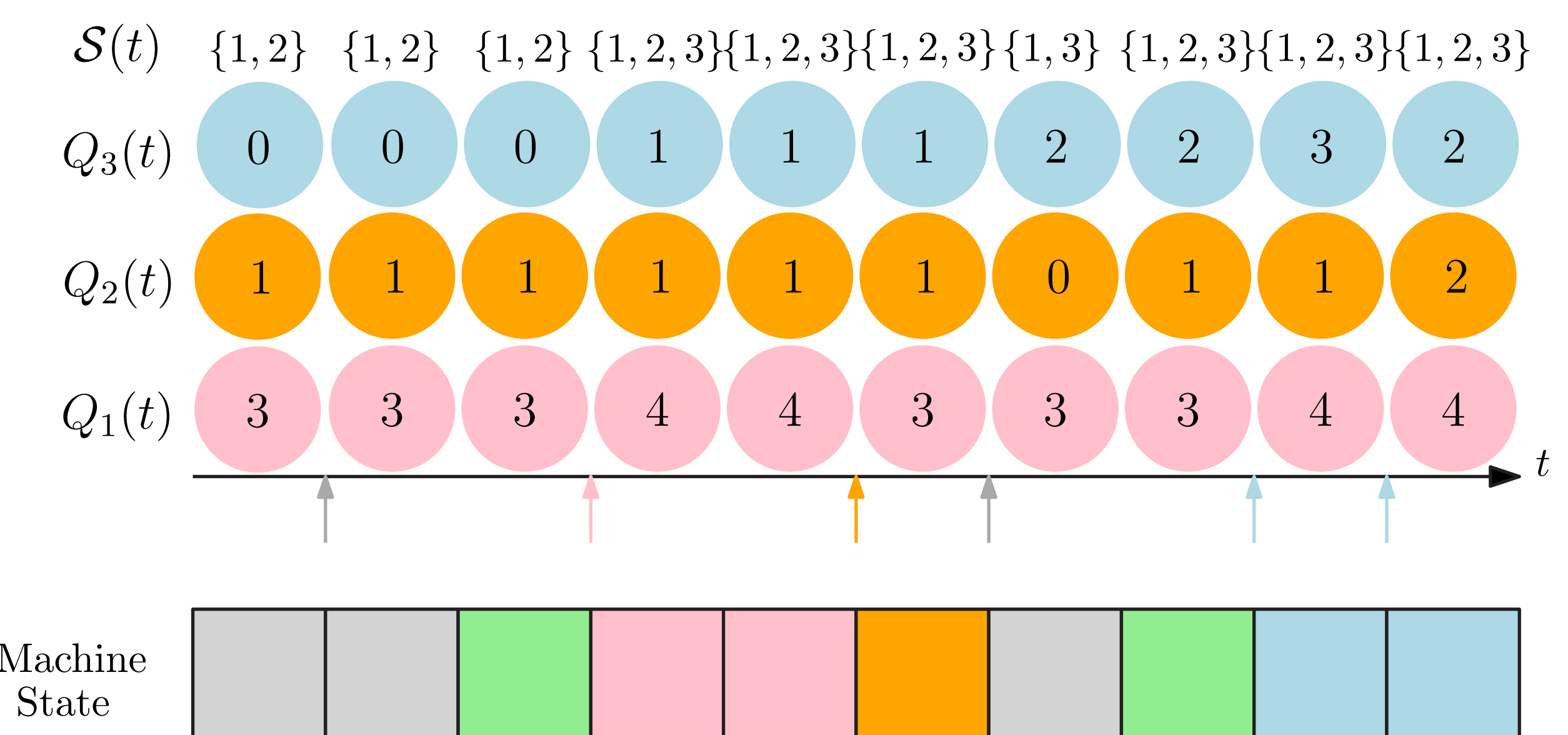
Take any subset of users $\mathcal{S} \neq \emptyset$ and assume their job queues remain non-empty $\forall t \geq 0$

Derive the average AoJC and sampling cost under stationary randomized sampling and scheduling policies

Optimize to obtain $(\mu^*(\mathcal{S}), \pi^*(\mathcal{S}))$

- At each time t , we observe the set of non-empty job queues $\mathcal{S}(t)$ and employ $(\mu^*(\mathcal{S}(t)), \pi^*(\mathcal{S}(t)))$.

Example:



Max-age scheduling policy with adaptive randomized sampling:

Take any subset of users $\mathcal{S} \neq \emptyset$ and assume their job queues remain non-empty $\forall t \geq 0$

Derive the average AoJC and sampling cost under stationary randomized sampling and max-AoJC scheduling policies

Optimize to obtain $(\bar{\mu}^*(\mathcal{S}), \pi^{MA}(\mathcal{S}))$

- At each time t , we observe the set of non-empty job queues $\mathcal{S}(t)$ and employ $(\bar{\mu}^*(\mathcal{S}(t)), \pi^{MA}(\mathcal{S}(t)))$.

7. Queue Stability

Proposition Under any adaptive randomized policy, the system is stable if $\exists \epsilon > 0$, such that

$$\sum_{i=1}^N p_i - \mu_{\min}(1 - \chi(q, s))q_{\min} \leq -\epsilon,$$

where $\mu_{\min} = \min_{\mathcal{S} \subseteq [N], \mathcal{S} \neq \emptyset} \{\mu(\mathcal{S})\}$ and $q_{\min} = \min_{i \in [N]} \{q_i\}$.

- We also find a sufficient condition for queue stability under the Max-age scheduling policy with adaptive randomized sampling.

8. Numerical Result

